Solution:

Marks in Commerce (X)	Rank (R_{1i})	Marks in Mathematics (Y)	Rank (R_{2i})	$D_i = R_{1i} - R_{2i}$	D_i^2
15	2	40	6	-4	16
20	3.5	30	4	-0.5	0.25
28	5	50	7	-2	4
12	1	30	4	-3	9
40	6	20	2	4	16
60	7	10	1	6	36
20	3.5	30	4	-0.5	0.25
80	8	60	8	0	0
				Total	$\sum D^2 = 81.5$

$$\rho = 1 - 6\left[\frac{\sum D_i^2 + \frac{1}{12}\left(m_1^3 - m_1\right) + \frac{1}{12}\left(m_2^3 - m_2\right) + \ldots}{n\left(n^2 - 1\right)}\right]$$

Repetitions of ranks

In Commerce (X), 20 is repeated two times corresponding to ranks 3 and 4. Therefore, 3.5 is assigned for rank 2 and 3 with m_1=2.

In Mathematics (Y), 30 is repeated three times corresponding to ranks 3, 4 and 5. Therefore, 4 is assigned for ranks 3,4 and 5 with m_2=3.

Therefore,

$$\rho = 1 - 6\left[\frac{81.5 + \frac{1}{12}\left(2^3 - 2\right) + \frac{1}{12}\left(3^3 - 3\right)}{8\left(8^2 - 1\right)}\right]$$

$$= 1 - 6\frac{[81.5 + 0.5 + 2]}{504} = 1 - \frac{504}{504} = 0$$

Interpretation: Marks in Commerce and Mathematics are uncorrelated

4.6 YULE'S COEFFICIENT OF ASSOCIATION



Udny yule

This measure is used to know the existence of relationship between the two attributes A and B (binary complementary variables). Examples of attributes are drinking, smoking, blindness, honesty, etc.

Udny Yule (1871 – 1951), was a British statistician. He was educated at Winchester College and at University College London. After a year dong research in experimental physics, he returned to University College in 1893 to work as a demonstrator for Karl Pearson. Pearson was beginning to work in statistics and Yule followed him into this new field. Yule was a prolific writer, and was active in Royal Statistical Society and received its Guy Medal in Gold in 1911, and served as its President in 1924–26.The concept of Association is due to him.

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